

04 — Control Plane

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Revision: 1 **Last modified:** 2026-07-05T14:20:00Z **Status:** Draft
— consolidated from 02-control-plane.md and v03-control-plane/*.

1. Position

The control plane is the **Go brain** that holds identity, topology, and policy truth, compiles ACLs, and pushes desired-state network maps to every edge. It **never sits in the packet path**.

Stack: Go + Gin + PostgreSQL + Redis + Podman (rootless).

2. Modular monolith layout

```
helix-go/
├── cmd/
│   ├── helixd/                # control-plane binary
│   └── helixvpctl/            # bootstrap + ops CLI
├── internal/
│   ├── identity/              # tenants, users, OIDC, anonymous
│   │   device tokens
│   │   ├── registry/          # devices, connectors, advertised
│   │   │   prefixes, presence
│   │   └── ipam/               # overlay IP allocation (ULA /48 per
│   │       tenant)
│   └── pki/                   # short-lived device certs, rotation,
│       revoke
│   ├── policy/                # ACL model + pure, versioned
│   └── compiler
│       ├── coordinator/        # in-mem topology graph, deltas,
│       │   WatchNetworkMap streams
│       └── events/             # bus abstraction (Redis Streams MVP
│           → NATS JetStream P2)
│       └── telemetry/          # Prometheus metrics + control-action
│           audit sink
```

		api/	# Gin REST + Connect (agents) + WS/
SSE			
		store/	# Postgres pool, RLS tenant-tx
helper, migrations			
		proto/	# helix/coordinator/v1/*.proto
		openapi/	# REST contract
		migrations/	# goose SQL – schema authority

3. Wiring rules

- **R1** — no cross-store imports; modules read each other through exported interfaces.
- **R2** — coordinator owns no durable tables; only in-memory graph + streams.
- **R3** — every state mutation emits an event (transactional outbox pattern).
- **R4** — every DB access is tenant-scoped under RLS.

4. Data model

Core tenant-scoped tables:

- tenants, users (with user_role enum: admin/operator/member)
- devices, connectors, advertised_prefixes
- groups, group_members
- policies (jsonb spec, monotonic version, active flag)
- overlay_pools (ULA /48 per tenant)
- device_certs (short-lived, revocable)
- audit_events (control actions only — never traffic)
- outbox (transactional event staging)

Absent by design: no connections, sessions, flows, traffic, packets tables. Their absence is CI-lint-enforced.

5. Row-Level Security

```
ALTER TABLE devices ENABLE ROW LEVEL SECURITY;
ALTER TABLE devices FORCE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON devices
  USING (tenant_id = current_setting('app.tenant_id')::uuid);
```

The app connects as a non-superuser helix_app role; tenant isolation is enforced at the database even if app-layer RBAC is bypassed.

6. Policy service and compiler (with GAP-1 consolidation)

The policy package is a **pure, deterministic, versioned compiler**:

- Input: Tailscale-ACL-flavored JSON document (`policies.spec`).
- Output:
 - `VisibleTo` — need-to-know peer set.
 - `AllowedIPs` — coarse WG routes.
 - `Verdicts` — port/proto-granular edge map.
 - `Via6` — 4via6 mappings for overlapping IPv4 LANs.
 - `ExitNodes` — full-tunnel grants.

GAP-1 — connector local-ACL precedence (adopted in consolidation):

The policy compiler accepts an optional `connector.local_denylist` input. The precedence rule is:

1. Local-deny overrides central-allow.
2. Central-deny overrides local-allow.
3. Output = central policy minus local-deny.

Honesty note: backported into `v03-control-plane/svc-policy.md` and `v04-client/helix-core-rust.md`; GAP-1 CLOSED.

7. Event backbone

- **MVP:** Redis Streams.
- **Phase 2:** NATS JetStream (mechanical swap via Bus interface).

Key streams: `device.enrolled`, `device.revoked`, `route.advertised`, `policy.updated`, `policy.compiled`.

Reliable publish via transactional outbox + sweeper.

8. Coordinator and WatchNetworkMap

The coordinator maintains an **in-memory per-tenant topology graph** and streams:

- Snapshot — full desired-state on first connect or version gap.
- Delta — minimal change set for the affected agent.

Peers are **pre-policy-filtered** before they hit the wire (need-to-know).

SLOs:

- Convergence p99 < 1 s (event → delta-on-wire).
- Revoke < 1 s (peer dropped everywhere).

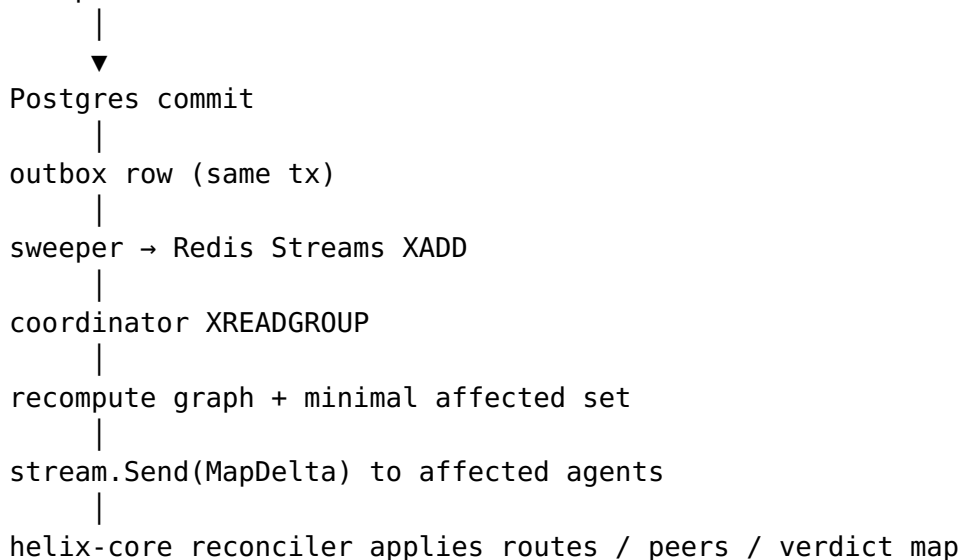
9. API surface

- **REST (Gin):** admin / Console CRUD, OpenAPI-generated clients.
- **Connect-RPC / gRPC:** agent WatchNetworkMap and enrollment.
- **WebSocket / SSE:** live Console state feed.

Auth: OIDC session or API token + RBAC (admin / operator / member), with RLS as the backstop.

10. Reconciliation flow

Admin/Operator → REST/CLI



11. Cross-references

- Data plane → [03 — Data Plane](#)
 - API contracts → [08 — API Contracts](#)
 - Security/PKI → [06 — Security, Privacy & PKI](#)
 - Client core → [05 — Client Core & UI](#)
 - Detailed DDL → [../v03-control-plane/data-model-ddl.md](#)
 - Detailed protobuf → [../v03-control-plane/protobuf-spec.md](#)
 - Detailed coordinator → [../v03-control-plane/svc-coordinator.md](#)
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Sources: docs/research/mvp/final/02-control-plane.md, v03-control-plane/.md.*